# NULL FUNCTIONS (ADVANCED SQL)

Q1 Select employee_id, Name
IFNULL (salary, 0) As Salary_with_default
FROM EMPLOYEES;

EMPLOYEES

| Employee_id | Name | Salary_with_default |
|---|---|---|
| 1 | ALICE | 50000 |
| 2 | BOB | 0 |
| 3 | Charlie | 7000 |
| 4 | DANA | 0 |

Q2. Select Employee_id, name
IFNULL (Department, 'Not Assigned') AS department_Name
FROM EMPLOYEES;

EMPLOYEES

| EMPLOYEE_ID | NAME | DEPARTMENT_NAME |
|---|---|---|
| 1 | ALICE | HR |
| 2 | BOB | IT |
| 3 | Charlie | NOT Assigned |
| 4 | DANA | FINANCE |

Q3 SELECT Order_id, CUSTOMER_id
FROM ORDERS
Where Customer_id IS NULL;

ORDERS

| ORDER_ID | CUSTOMER_ID |
|---|---|
| 103 | NULL |

Q4. Select Order_ID, Customer_id,
[illegible] 'Pending') AS delivery_status

IFNULL (delivery_date, 'PENDING')
FROM ORDERS;

ORDERS

| ORDER_ID | CUSTOMER_ID | DELIVERY_STATUS |
|---|---|---|
| 101 | 201 | 2024-12-01 |
| 102 | 202 | PENDING |
| 103 | NULL | 2024-12-03 |

Q5 SELECT STUDENT_ID, NAME
IFNULL (GRADE, 0) AS FINAL_GRADE
FROM STUDENTS;

STUDENTS

| STUDENT_ID | NAME | FINAL_GRADE |
|---|---|---|
| 1 | ETHAN | 85 |
| 2 | MAYA | 0 |
| 3 | OLIVIA | 90 |

~~Q6~~ SELECT ~~PRODUCT_ID, NAME~~
~~PRICE~~

Q6. SELECT COUNT(*) AS NOT_GRADED_COUNT
FROM STUDENTS
WHERE GRADE IS NULL;

STUDENTS

| NOT_GRADED_COUNT |
|---|
| 1 |

Q7. SELECT PRODUCT_ID, NAME
COALESCE (discount, 0) AS Final_Price
FROM Products;

PRODUCTS

| Product_id | NAME | FINAL_PRICE |
|---|---|---|
| S01 | KEYBOARD | 25 |
| S02 | MOUSE | 10 |
| S03 | MONITOR | 100 |

Q8. SELECT (*) AS MISSING_EMAIL_COUNT
FROM CUSTOMERS
Where email IS NULL;

CUSTOMERS

| Missing_email_count |
|---|
| 2 |

Q9. Select CUSTOMER_ID, NAME,
IFNULL (Email, 'NO EMAIL') AS email_display
FROM CUSTOMERS;

CUSTOMERS

| CUST_ID | NAME | EMAIL_Display |
|---|---|---|
| 1 | LINDA | NO EMAIL |
| 2 | JOSEPH | Joseph@mail.com |
| 3 | NIA | NO EMAIL |

Q10. Select payment_id,
IFNULL (Method, 'UNKOWN') AS Method_display,
STATUS
FROM PAYMENTS;

PAYMENTS

| PAYMENT_ID | METHOD_DISPLAY | STATUS |
|---|---|---|
| 301 | CREDIT | NULL |
| 302 | PAYPAL | Success |
| 303 | UNKOWN | FAILED |

Q11. Select item_id, ITEM_Name
IFNULL (Quantity, 0) AS quantity_checked
FROM INVENTORY;

INVENTORY

| ITEM_ID | ITEM_NAME | Quantity_Checked |
|---|---|---|
| 1 | PEN | 0 |
| 2 | NOTEBOOK | 150 |
| 3 | ERASER | 0 |

Q12. Select emp_id
Coalesce (Bonus, commission, 0) As first_Available_Reward
From Employees_EXTRA;

EMPLOYEES_EXTRA

| EMP_id | FIRST_Available_Reward |
|---|---|
| 1 | 300 |
| 2 | 100 |
| 3 | 0 |

Q13. Select Count(*) AS NO_ROOM_COUNT
From Classes
Where Room IS NULL;

Classes

| NO_ROOM_COUNT |
|---|
| 2 |

Q14. Select Student_id, date
IFNULL (STATUS, 'NOT Marked') AS ATTendance_STATUS
FROM ATTENDANCE;

ATTENDANCE

| STUDENT_ID | DATE | ATTENDANCE_STATUS |
|---|---|---|
| 1 | 2025-04-01 | NOT MARKED |
| 2 | 2025-04-01 | PRESENT |
| 3 | 2025-04-01 | ABSENT |

Q15. SELECT ACCOUNT_ID
IFNULL (ACCOUNT_TYPE, 'UNKOWN') AS TYPE_DISPLAY
IFNULL (BALANCE, 0) AS BALANCE_checked
FROM BANK_ACCOUNTS;

BANK_ACCOUNTS

| ACCOUNT_ID | TYPE_DISPLAY | BALANCE_Checked |
|---|---|---|
| A1 | SAVINGS | 0 |
| A2 | CURRENT | 5000 |
| A3 | UNKOWN | 2000 |

Q16. Select PROJECT_ID, TITLE
IFNULL (START_DATE, 'TBD') AS START_DISPLAY
FROM PROJECTS;

PROJECTS

| PROJECT_ID | TITLE | START_DISPLAY |
|---|---|---|
| 1 | Website Revamp | 2025-01-10 |
| 2 | Mobile APP | TBD |
| 3 | DATA Migration | TBD |

Q17. Select Review_ID, Product_ID,
IFNULL (Comment, 'NO COMMENT') AS Comment_display
IFNULL (Rating, 0) AS Rating_display
FROM REVIEWS;

Reviews

| Review_id | Product_id | Comment_display | Rating_display |
|---|---|---|---|
| 1 | S01 | Great product | 4 |
| 2 | S02 | No Comment | 0 |
| 3 | S03 | Works fine | 3 |

Q18. Select supplier_id, name
Coalesce (Phone, alt_phone, 0) AS Contact_NUMBER
FROM Suppliers;

Suppliers

| Supplier_id | name | Contact_Number |
|---|---|---|
| 1 | Global Goods | 123456789 |
| 2 | Best suppliers | 987654321 |
| 3 | Value source | 0 |

Q19. Select user-id
IFNULL (Theme, 'Light') AS Theme_set,
IFNULL (Language, 'ENGLISH') AS Language_set,
IFNULL (Timezone, 'UTC') AS timezone_set
FROM USER_SETTINGS;

USER_SETTINGS

| USER_ID | THEME_SET | Language_set | timezone_set |
|---|---|---|---|
| 1 | Light | English | UTC |
| 2 | DARK | English | UTC+1 |
| 3 | Light | English | UTC |

Q20. Select Record_id, Machine_id,
IFNULL (ISSUE, 'UNKOWN ISSUE') AS Issue_log,
IFNULL (Technician, 'NOT Assigned') AS technician_name
FROM Maintanance;

Maintanance

| Record_id, | Machine_id | ISSUE_log | Technician_Name |
|---|---|---|---|
| 1 | M101 | Overheating | NOT Assigned |
| 2 | M102 | Unkown issue | NOT Assigned |
| 3 | M103 | Jammed | Alex |